



**TECHNO
INTERNATIONAL
BATANAGAR**

A unit of  **Techno India Group**

DIVIDE & CONQUER METHOD

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Branch : Computer Science & Engineering

Subject : Artificial Intelligence

INTRODUCTION

A divide-and-conquer algorithm recursively breaks down a problem into two or more sub-problems of the same or related type, until these become simple enough to be solved directly. The solutions to the sub-problems are then combined to give a solution to the original problem.

A typical Divide and Conquer algorithm solves a problem using following three steps.

- Divide: Break the given problem into subproblems of same type.
- Conquer: Recursively solve these subproblems.
- Combine: Appropriately combine the answers

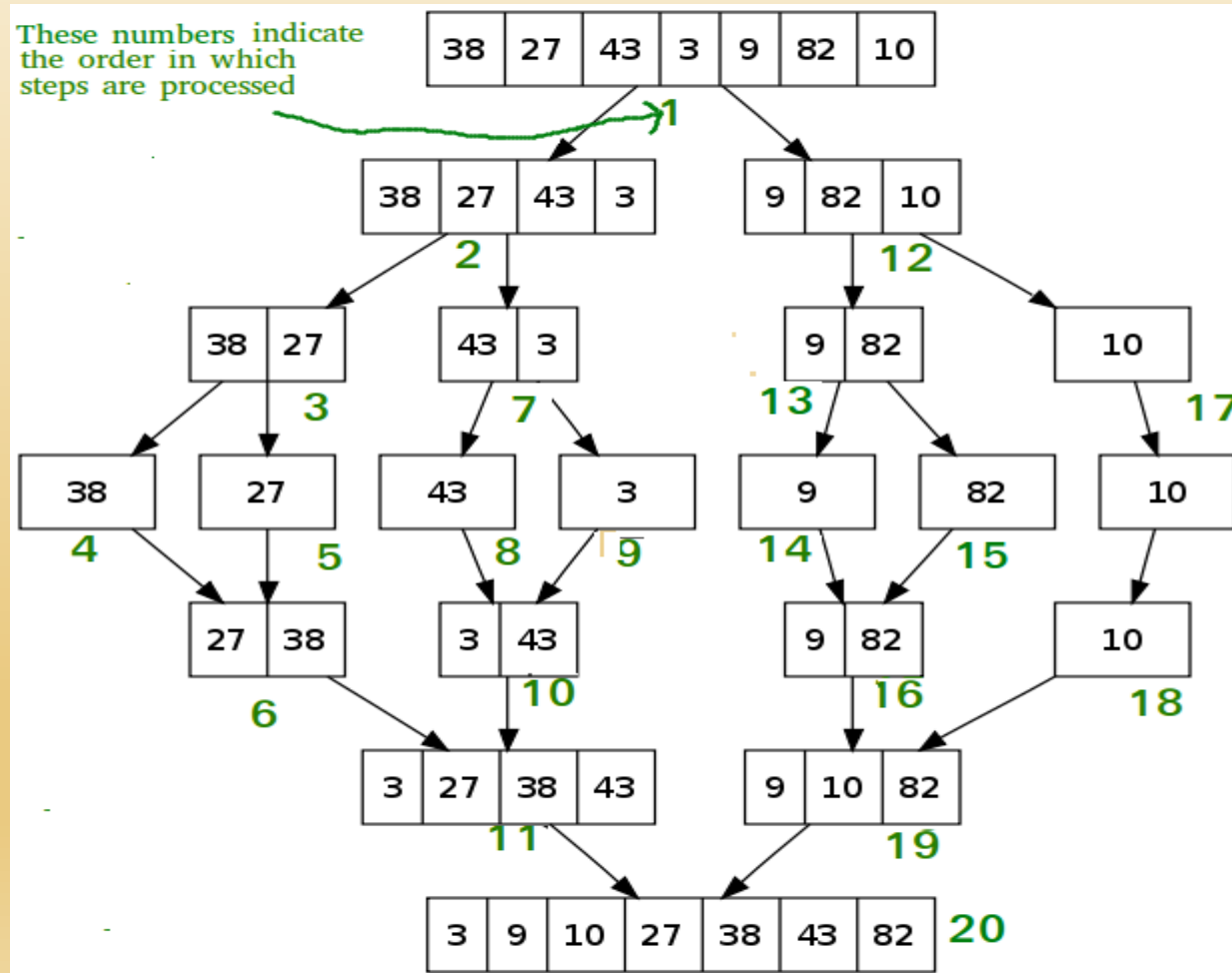
The Following Are Some Standard Algorithms That Follow Divide and Conquer Algorithm

- Quick Sort.
- Merge Sort.
- Binary Search.
- Find Maximum & Minimum Number.
- Strassen's Algorithm.

Divide & Conquer Algorithm

```
DAC(a, i, j)  
{  
    if(small(a, i, j))  
        return(Solution(a, i, j))  
    else  
        m = divide(a, i, j)  
        b = DAC(a, i, mid)  
        c = DAC(a, mid+1, j)  
        d = combine(b, c)  
    return(d)  
}
```

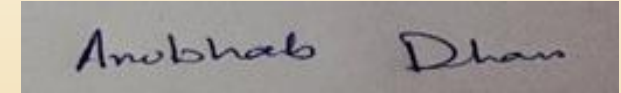
How Divide & Conquer Algorithms Work



Declaration:

I hereby declare that this ppt is made by me only and not downloaded or copied from any other sources.

Date: 24.07.2022

A rectangular box containing a handwritten signature in dark ink. The signature appears to read "Anubhab Dhar".

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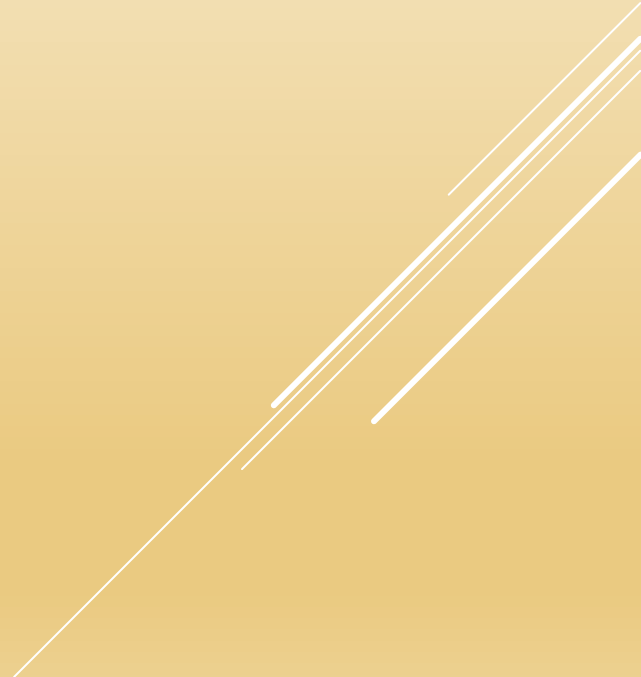
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Thank You

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